

powered, both LED12 and LED13 remain off. When switch S9 is pressed, LED12 turns on, and when S10 is pressed, LED12 turns off. Similarly, when S11 is pressed, LED13 turns on, and when S12 is pressed, LED13 turns off. This is a normal bistable operation.

But if switch S9 is pressed to switch on LED12 and then S11 is pressed, instead of LED13 turning on LED12 turns off. Thus, both LED12 and LED13 remain off though the output at pin 3 of both the ICs is high! This unique situation arises because the internal discharge transistors at pin 7 of both the ICs turn off and the transistors cannot conduct.

Now, in this situation, if S10 is pressed, LED13 turns on, and its on and off operation can be controlled by S9 and S10. Likewise, the on and off operation of LED12 can be

controlled by S11 and S12. However, both the LEDs can never be on together.

Experiment 8

Circuit diagram for the eighth experiment, shown in Fig. 8, has two parts: 8(a) and 8(b).

In part 8(a) both the ICs are independently wired as astable multivibrators. They are identical, except the value of C5 is lower than that of C6. Normally, if the ICs are powered separately, both LED14 and LED15 flash, but the flashing of LED14 is faster than that of LED15.

However, in this circuit diode D1 is connected between pin 7 of IC14 and capacitor C6. So, when the circuit is powered, LED14 flashes but LED15 stays on continuously. This is because capacitor C6 keeps discharging through pin 7 of IC14 through diode D1 and astable

operation of IC15 fails to occur.

Now, if the orientation of diode D1 is reversed (cathode of diode connected to pin 7 of IC15 and anode to the positive end of capacitor C5), both LED14 and LED15 start flashing, though LED14 flashes at a slower rate as capacitor C5 intermittently discharges through pin 7 of IC15.

In part 8(b) the two ICs are again independently wired as astable multivibrators. The output from pin 3 of both the ICs is commonly connected to LED16 through diodes D2 and D3. When the circuit is powered, LED16 'seems' to glow continuously. However, in fact, LED16 is powered alternately by two independent astable multivibrators continuously.

When on/off switch S13 is closed, LED16 turns off as the circuit resets. If it is opened, LED16 glows immediately as the astable multivibrator operation is self-triggering. So, the circuit with two independent astable multivibrators together 'appears' to act as a bistable multivibrator!

Experiment 9

As shown in Fig. 9, in this experiment both the ICs are once again wired as astable multivibrators. They are identical, except capacitor C9 is of much lower value than capacitor C10. When powered independently, LED17 flashes fast

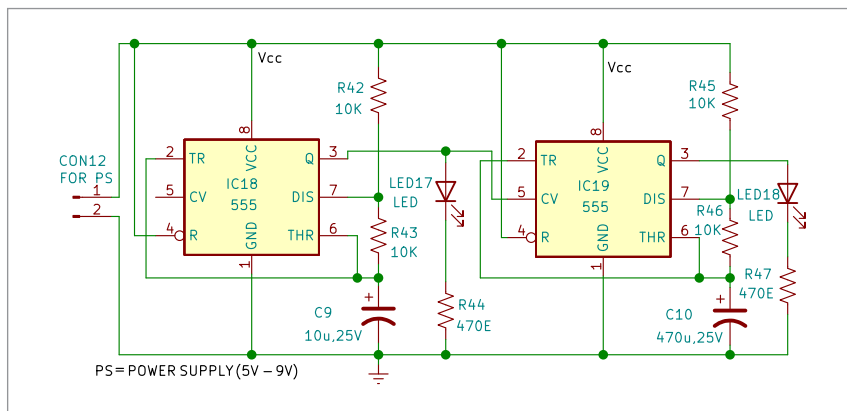


Fig. 9: Circuit diagram for experiment 9

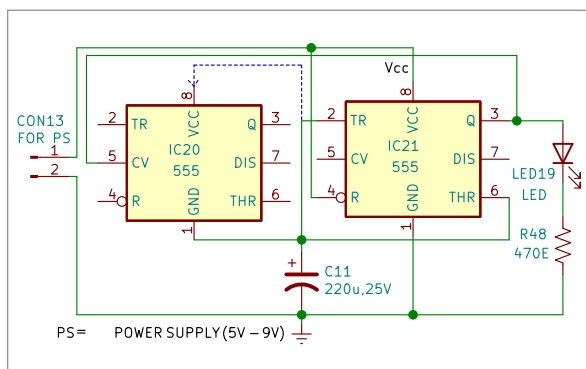


Fig. 10: Circuit diagram for experiment 10

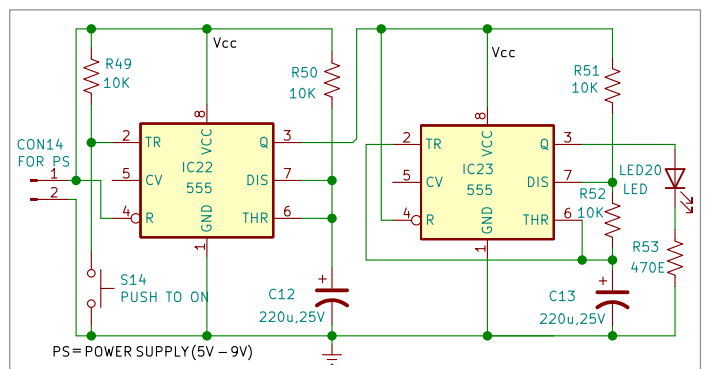


Fig. 11: Circuit diagram for experiment 11